

EI240 FIRE PERFORMANCE TECHNICAL DATASHEET

High-Density Basaltic Rock Wool Sandwich Panels

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Certification Authority: Fire Safety Institute System Reference: Rockfire-240 Panel

1 Applicable Standards and Certification

The Rockfire-240 sandwich panel system has undergone rigorous fire resistance testing, achieving the highest fire performance ratings according to current European and international standards:

- **EN 13501-2:** Fire classification of construction products and building elements – Part 2: Classification using data from fire resistance tests, excluding ventilation services.
- **EN 1364-1:** Fire resistance tests for non-loadbearing elements – Part 1: Walls.

2 Understanding the EI240 Classification

The certified fire resistance rating of **EI240** defines the exact physical and thermal performance of the building assembly when exposed to a fully developed fire scenario:

- **E (Integrity):** The ability of the panel system to prevent the passage of flames and hot, ignitable gases to the unexposed face for at least 240 minutes, successfully containing the physical spread of fire.
- **I (Insulation):** The ability to restrict heat transmission via conduction and radiation. The system guarantees that the average temperature rise on the unexposed face is below 140°C, and that local hot spots do not exceed a rise of 180°C for a minimum duration of 240 minutes.
- **240:** The guaranteed minimum execution period in minutes ($\geq$ 240 minutes) during which both criteria (Integrity and Insulation) are strictly met.

3 System Construction Specifications

To ensure certified EI240 performance, the construction assembly must be strictly installed in accordance with the following tested configuration:

System Parameter	Certified Technical Specification
Nominal Panel Thickness	$\geq$ 150 mm
Insulating Core Density	135 kg/m ³ (Oriented fiber basaltic rock wool core)
Metallic Facing Sheets	0.6 mm thick galvanized and pre-painted steel cladding
Joint Geometry	Double tongue-and-groove joint with high-grade intumescent sealant
Supporting Frame	Cold-formed structural steel profiles
Fastener Assembly	M8 heavy-duty mechanical anchors with neoprene sealing washers

4 Performance in Testing under ISO 834 Standard Curve

During the official fire testing procedure, the panel division wall was exposed to the thermal gradient defined by the standard ISO 834 curve. The main chronological milestones recorded during the test were:

- **Minute 0:** Burners ignited and thermal gradient ramp started following the ISO 834 standard curve.
- **Minute 60 (945°C):** Complete structural stability. Zero structural deflection or lateral buckling observed in the partitioning panel system.
- **Minute 120 (1049°C):** Insulation performance (I) remains flawless. The maximum temperature rise on the unexposed face registers below 45°C.
- **Minute 180 (1110°C):** The high-density basaltic rock wool core actively blocks extreme thermal radiation. Joint intumescent sealant swells to form an air-tight, hot-gas barrier.
- **Minute 240 (1153°C):** Successful completion of the official test. No unexposed surface sensors exceeded the strict 180°C local threshold limit. Zero integrity failure, cracking, or flame penetration detected (E).

5 Applications and Regulatory Compliance

The Rockfire-240 high-performance panel is engineered as an advanced structural solution for high-security fire compartmentalization in high-risk zones:

- **Fire Compartmentation Walls:** Secure dividing barriers in large-scale logistics warehouses and manufacturing facilities to isolate distinct industrial hazard sectors.
- **Critical Equipment Enclosures:** Enclosing structures for boiler rooms, high-voltage substations, and critical data processing facilities (data centers).
- **Protected Escape Routes:** Partitioning walls for emergency escape paths and public corridors where thermal radiation prevention and absolute containment of contiguous fires are vital.